



Russian Internet

Russia has commenced testing a nation-wide internet that would be separate from the internet that we know <https://dgit.in/jan20-47>



Auto calibrating speakers

LG's 2020 soundbars will reportedly have AI-enabled room calibration to adjust the audio signature <https://dgit.in/jan20-48>

MSI Creator X299

Enriching creator workflow

MSI motherboards have been knocking it out of the park over these last couple of generations. A recent example would be with the X399 Threadripper platform wherein MSI's MEG X399 Creation ended up being one of the most feature rich and targeted motherboards than money can buy. For the X299 refresh, MSI is hoping for another home run with the Creator X299.

Board Layout

The Intel X299 platform has scope for incorporating a lot of features given the sheer number of PCIe lanes that the chipset as well as the Extreme Edition Intel processors have available. So it makes sense to use an E-ATX motherboard layout which has plenty of real estate to incorporate as many features as possible.

Starting with the VRM, the Creator X299 has a 12 phase VRM with 90A power stages. That's right, 90A. Usually, we see 60-70A power stages but this board is on a different level altogether considering that this setup can provide more than enough power to handle aggressive overlocks. The Extreme Edition processors are certainly power hogs but not to this level. We've seen the 10980XE consume a little over 200 watts while loaded and without overclock. This board has three 8-pin EPS connectors which means it can theoretically draw 336 watts per connector so that's a total of 1008 watts. However, things aren't so linear and you're going to see the processor consume way below what the three connectors are capable of providing. Moreover, continuous power for EPS connectors is around 235 watts. Having the load spread across three EPS connectors reduces the heat generated under high load conditions. MSI is known to use IR3598 or IR3599 dou-



Price
₹60,000
(Approx)



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blers in most of their designs but they've supposedly done away with that here.

On the VRMs we have a massive aluminium heat-sink which even has a heat pipe embedded in it for extra cooling. There is a decent amount of clearance around the CPU socket (59 mm on either side) and the VRM heatsink isn't outrageously tall to block the installation of certain CPU coolers which sport massive finstacks and 140 mm fans.

About half of the motherboard is hidden under the three M.2 heatsinks. Speaking of which, there is enough gap between the PCIe sockets to easily fit NVMe SSDs such as the WD Black SN750 that have wider heatsinks.

The SATA and USB 3.0 power connectors along the right edge use sideways-facing connectors except the main 24-pin ATX connector. The debug LED as well as the OC Genie knob are the the top-right corner which is easier to access if you have all four PCIe slots occupied. After all, this board does support 4-way SLI/Crossfire.

Lastly, the I/O shield comes attached to the board and incorporated plenty of USB 3.x ports and even two Ethernet ports. The audio jacks are all gold plated to prevent corrosion.

Key Features

This board is aimed at creators which means there has to be support for plenty of interfaces that are typically part of a creator's workflow and there should be the stability to ensure that the workflow remains as unhindered as possible. Since creators tend to work with massive files, be it for video editing or 3D modelling, being able to access these files quickly is essential. For this, MSI provides an M.2 XPANDER Aero card which allows you to connect four M.2 NVMe SSDs via the PCIe port and you can make use of Intel VROC to set up a virtual RAID. The X299 platform does not have the advantage of PCIe 4.0 but for most creator workflows, the speed provided by PCIe 3.0 should be more than enough. Unfortunately, VROC has always been a separate paid accessory. You can always use the M.2 Genie to set up RAID 0 on the three M.2 connectors that are on the motherboard. Then there's a Thunderbolt expansion card which has two Type-C Thunderbolt ports as well as two DisplayPorts. Then there's the inclusion of 10 Gig Ethernet connectivity thanks to the Aquantia AQC107 controller. This would certainly help with moving large files to and from your local NAS.

Verdict

The MSI Creator X299 is aimed at leveraging the recently refreshed X299 platform with a focus on enabling creators. The board comes with several features to this extent and the build quality is excellent. Indian pricing for this board should be around ₹60,000 which isn't that outrageous considering the number of features that this board has.

—Mithun Mohandas

SPECIFICATIONS

CHIPSET: Intel X299 | MEMORY SUPPORT: 4x4266(OC)MHz upto 256GB | LAN: Intel i219-V GbE, Aquantia 10G | AUDIO: Realtek ALC1220 | SATA: 8x SATA III | EXPANSION: 4x PCIe x16(x8/x8/x16/x8) with 48-lane CPU | DIMENSIONS (LxWxD): 305 mm x 272 mm | WARRANTY: 3 years.

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